

Databases I - CPIT240

LAB-2

SQL Data Definition Language

(Create Simple Table)

SQL Statement

- It is an instruction that consists of identifiers, parameters, variables, names, data types, and SQL reserved words.

SQL Statements Types

- Data Definition Language (**DDL**) Statements.
- Data Manipulation Language (**DML**) Statements.
- Transaction Control Language (**TCL**) Statements.

DDL	DML	TCL
CREATE	SELECT	COMMIT
ALTER	INSERT	ROLLBACK
DROP	UPDATE	SAVEPOINT
TRUNCATE	DELETE	
RENAME	MERGE	
GRANT		
REVOKE		
COMMENT		

Data Definition Language (DDL)

- It is for **creating** and **modifying** database objects.

CREATE

Create, alter, and drop schema objects ,the database itself, and database users.

ALTER

DROP

TRUNCATE

Delete all the data in schema objects without removing the structure of these objects.

RENAME

Set a new name for any existing table.

GRANT

Grant and revoke privileges and roles.

REVOKE

COMMENT

Add a comment about a table, view, or column into the data dictionary.

Data Types

- When you identify a column for a table, you need to provide a data type for the column.
- A datatype is associated with a specific storage format, constraints, and a valid range of values.
- There are several data types available:

Data Types

Data Type	Description
VARCHAR2(size)	Variable-length character data (A maximum size must be specified: 1 - 4,000 bytes or characters).
CHAR [(size)]	Fixed-length character data (1 - 2,000 bytes or characters)
NUMBER [(p,s)]	Number having precision p and scale s . p is the total number of digits (1 to 38) s is the number of digits to the right of the decimal point (-84 to 127)

Data Type	Description
DATE	Date and time values to the nearest second between January 1, 4712 B.C., and December 31, 9999 A.D.
LONG	Variable-length character data (up to 2 GB)
CLOB	Character data (up to 4 GB)
RAW(size)	Raw binary data of length size (A maximum size must be specified: maximum size is 2,000.)
LONG RAW	Raw binary data of variable length (up to 2 GB)
BLOB	Binary data (up to 4 GB)
BFILE	Binary data stored in an external file (up to 4 GB)
ROWID	A base-64 number system representing the unique address of a row in its table

Table 26-1 How Scale Factors Affect Numeric Data Storage

Input Data	Specified As	Stored As
7,456,123.89	NUMBER	7456123.89
7,456,123.89	NUMBER (*, 1)	7456123.9
7,456,123.89	NUMBER (9)	7456124
7,456,123.89	NUMBER (9, 2)	7456123.89
7,456,123.89	NUMBER (9, 1)	7456123.9
7,456,123.89	NUMBER (6)	(not accepted, exceeds precision)
7,456,123.89	NUMBER (7, -2)	7456100

If you specify a [negative scale](#), Oracle Database rounds the actual data to the specified number of places to the left of the decimal point. [For example](#), specifying (7,-2) means Oracle Database rounds to the nearest hundredths.

SYSDATE Function

- **SYSDATE** is a date function that returns the current database server date and time.
- You can use SYSDATE just as you would use any other column name.
- It is customary to **select SYSDATE** from a dummy table called **DUAL**.

```
SQL> select sysdate from dual;
```

```
SYSDATE
```

```
-----
```

```
03-FEB-21
```

Oracle DUAL table

- **DUAL** is a table automatically created by Oracle Database.
- It is in the schema of the user **SYS** but is accessible by the name DUAL to **all users**.
- It has one column, **DUMMY**, defined to be **VARCHAR2(1)**, and contains one row with a value **X**.
- It is used for **evaluating expressions** or **calling functions**.

```
SQL> select * from dual;
```

```
D  
-  
X
```

```
SQL> select 1+2 from dual;
```

```
      1+2  
-----  
      3
```

Tables Used in this Part

- **EMPLOYEES table:** Gives details of all the employees.
- **DEPARTMENTS table:** Gives details of all the departments.
- **JOB_GRADES table:** Gives details of salaries for various grades.

EMPLOYEES

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
100	Steven	King	SKING	515.123.4567	06/17/1987	AD_PRES	24000	-	-	90
101	Neena	Kochhar	NKOCHHAR	515.123.4568	09/21/1989	AD_VP	17000	-	100	90
102	Lex	De Haan	LDEHAAN	515.123.4569	01/13/1993	AD_VP	17000	-	100	90
103	Alexander	Hunold	AHUNOLD	590.423.4567	01/03/1990	IT_PROG	9000	-	102	60
104	Bruce	Ernst	BERNST	590.423.4568	05/21/1991	IT_PROG	6000	-	103	60
107	Diana	Lorentz	DLORENTZ	590.423.5567	02/07/1999	IT_PROG	4200	-	103	60
124	Kevin	Mourgos	KMOURGOS	650.123.5234	11/16/1999	ST_MAN	5800	-	100	50
141	Trenna	Rajs	TRAJS	650.121.8009	10/17/1995	ST_CLERK	3500	-	124	50
142	Curtis	Davies	CDAVIES	650.121.2994	01/29/1997	ST_CLERK	3100	-	124	50
143	Randall	Matos	RMATOS	650.121.2874	03/15/1998	ST_CLERK	2600	-	124	50
144	Peter	Vargas	PVARGAS	650.121.2004	07/09/1998	ST_CLERK	2500	-	124	50
149	Eleni	Zlotkey	EZLOTKEY	011.44.1344.429018	01/29/2000	SA_MAN	10500	.2	100	80
174	Ellen	Abel	EABEL	011.44.1644.429267	05/11/1996	SA_REP	11000	.3	149	80
176	Jonathon	Taylor	JTAYLOR	011.44.1644.429265	03/24/1998	SA_REP	8600	.2	149	80
178	Kimberely	Grant	KGRANT	011.44.1644.429263	05/24/1999	SA_REP	7000	.15	149	-
										row(s) 1 - 15 of 20 >

DEPARTMENTS

DEPARTMENT_ID	DEPARTMENT_NAME	MANAGER_ID	LOCATION_ID
10	Administration	200	1700
20	Marketing	201	1800
50	Shipping	124	1500
60	IT	103	1400
80	Sales	149	2500
90	Executive	100	1700
110	Accounting	205	1700
190	Contracting	-	1700
			row(s) 1 - 8 of 8

JOB_GRADES

GRADE_LEVEL	LOWEST_SAL	HIGHEST_SAL
A	1000	2999
B	3000	5999
C	6000	9999
D	10000	14999
E	15000	24999
F	25000	40000
		row(s) 1 - 6 of 6

Tables Used in this Part

EMPLOYEES

Column Name	Data Type	Nullable	Default	Primary Key
EMPLOYEE_ID	NUMBER(6,0)	No	-	1
FIRST_NAME	VARCHAR2(20)	Yes	-	-
LAST_NAME	VARCHAR2(25)	No	-	-
EMAIL	VARCHAR2(25)	No	-	-
PHONE_NUMBER	VARCHAR2(20)	Yes	-	-
HIRE_DATE	DATE	No	-	-
JOB_ID	VARCHAR2(10)	No	-	-
SALARY	NUMBER(8,2)	Yes	-	-
COMMISSION_PCT	NUMBER(2,2)	Yes	-	-
MANAGER_ID	NUMBER(6,0)	Yes	-	-
DEPARTMENT_ID	NUMBER(4,0)	Yes	-	-
				1 - 11

DEPARTMENTS

Column Name	Data Type	Nullable	Default	Primary Key
DEPARTMENT_ID	NUMBER(4,0)	No	-	1
DEPARTMENT_NAME	VARCHAR2(30)	No	-	-
MANAGER_ID	NUMBER(6,0)	Yes	-	-
LOCATION_ID	NUMBER(4,0)	Yes	-	-
				1 - 4

JOB_GRADES

Column Name	Data Type	Nullable	Default	Primary Key
GRADE_LEVEL	VARCHAR2(3)	Yes	-	-
LOWEST_SAL	NUMBER	Yes	-	-
HIGHEST_SAL	NUMBER	Yes	-	-
				1 - 3

How to create a table

SQL STATEMENT 1

- `CREATE TABLE tablename (column datatype [DEFAULT expr][, ...]);`

In the syntax:

- `tablename` is the name of the table.
- `DEFAULT expr` Specifies a default value if a value is omitted in the INSERT Statement (to be covered later).
- `column` Is the name of the column.
- `datatype` Is the column's data type and length.

DEFAULT Option

- When you define a table, you can specify that a column be given a **default value** by using the **DEFAULT** option.
- This option **prevents null values** from entering the columns.

Naming Rules

You name database tables and columns according to the standard rules for naming any Oracle database object:

- Table names and column names **must begin with a letter** and be **1–30 characters long**.
- Names must contain only the characters **A–Z, a–z, 0–9, _** (underscore), **\$**, and **#** (legal characters, but their use is discouraged).
- Names must **not duplicate** the name of another object owned by the same Oracle server user.
- Names must **not be an Oracle server–reserved word**.

Naming Guidelines

- Use **descriptive names** for tables and other database objects.
- **NOTE:** Names are **case-insensitive**.

For example: **EMPLOYEES** is treated as the same name as **employees**.

TRY THIS

TRY THIS

1. Log in to SQL*Plus and type the following

```
CREATE TABLE dept  
(deptno      NUMBER(2) ,  
  dname      VARCHAR2(14) ,  
  loc        VARCHAR2(13) ,  
  create_date DATE DEFAULT SYSDATE) ;
```

This creates the **DEPT** table, with four columns: **DEPTNO**, **DNAME**, **LOC**, and **CREATE_DATE**. The **CREATE_DATE** column has a default value. If a value is not provided when inserting data, the system date is automatically inserted.

2. Confirm that the table has been created:
DESCRIBE dept

How to Alter a table

- After you create a table, you may need to **modify** the table structure.
- You can do this by using the **ALTER TABLE** statement.
- Use the **ALTER TABLE** statement to:
 - **Add** a new column.
 - **Modify** an existing column.
 - **Define** a default value for the new column.
 - **Drop** a column.

How to Alter a table

- **Adding** a New Column or define a default value for a column

```
ALTER TABLE tablename ADD (columnName datatype [DEFAULT expr][, ...]);
```

- **Modify** an existing column or define a default value for a column

```
ALTER TABLE tablename MODIFY (columnName datatype [DEFAULT expr][, ...]);
```

- **Drop** a column

```
ALTER TABLE tablename DROP (columnName);
```

TRY THIS

TRY THIS

1. Type the following to add a column:

```
ALTER TABLE dept  
ADD (job_id VARCHAR2(9));
```

2. Confirm that the column has been added:

```
DESCRIBE dept
```

TRY THIS

TRY THIS

1. Type the following to modify a column:

```
ALTER TABLE dept  
MODIFY (loc VARCHAR2(30));
```

2. Confirm that the column has been modified:

```
DESCRIBE dept
```

TRY THIS

TRY THIS

1. Type the following to drop a column:

```
ALTER TABLE dept
```

```
DROP COLUMN job_id;
```

2. Confirm that the column has been dropped:

```
DESCRIBE dept
```

How to Drop a table

- DROP TABLE tablename.

TRY THIS

1. Type the following to drop a table:
`DROP TABLE dept;`
2. Confirm that the table has been dropped:
`DESCRIBE dept`

How to Truncate a table

- The SQL statement **removes all rows** from a table, leaving the table empty and the table structure **intact**.
- **TRUNCATE TABLE** **tablename**;

How to rename a table

- **RENAME** old tablename **TO** new tablename;

TRY THIS

1. Check that the departments table exists:
`DESCRIBE dept;`
2. Rename the departments table to emp_departments:
`RENAME dept TO emp_departments;`
3. Check the structure of the emp_departments table:
`DESCRIBE emp_departments;`
4. Rename the emp_departments table to departments again:
`RENAME emp_departments TO dept;`